Kevin Benny Mathew

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Education

2023 - 2027	B.E. (Hons.) Electronics and Communication Engineering - BITS Pilani, Goa Campus	**8.21/10** **(Upto Semester 4)**
May 2023	ISC (Class XII) - The Bishop's School, Camp	**97.6%**
May 2021	ICSE (Class X) - St. Mary's School, Pune	**99.0%**

Skills & Proficiency

Languages C++, C, Python, Java, Verilog
Technologies Linux, ROS/ROS2, Microcontrollers (ESP32, Teensy, Arduino), Git
Key Courses Microprocessors & Interfacing, Digital Signal Processing, Object-Oriented Programming, Digital Design, VLSI Design, Communication Systems, Operating Systems (Ongoing), Computer Architecture (Ongoing)
Tools Vivado, Cadence Virtuoso, PlatformIO, MATLAB, Android Studio, KiCAD, Fusion 360

Work Experience

Summer Intern May 2025 - Aug 2025

- Worked at the **Indian Meteorological Department** on sensor fusion and embedded software development.
- Developed embedded firmware for ESP8266 hardware using C++ and PlatformIO.
- Implemented custom fuzzy logic algorithms and Kalman filters to improve sensor accuracy.
- Conducted data analysis using Python and documented software performance for research.

Projects

Project Kratos: Mars Rover (Electronics Subsystem) Aug 2024 - Sep 2025

- Migrated the entire robot code stack from ROS Noetic to ROS2 Humble to improve real-time performance.
- Redesigned the electronics architecture for modularity and improved system-level debugging.
- Core member of the electronics subsystem responsible for hardware-software integration.

Low Cost Electromechanical Tensile Tester May 2025 - Dec 2025

- Integrated motion control firmware using Teensy microcontrollers and stepper motors.
- Developed a custom user interface (UI) to control the testing parameters and visualize data.
- Executed end-to-end development including electronics architecture and mechanical design.

Project Vulcan: Humanoid Robot Feb 2024 - Present

- Led the bionic arm subsystem, developing control logic for a 7-DOF biomimetic robotic arm.
- Implemented audio signal processing circuits for microphones located in the robot's ears.
- Improved internal mechanical systems for facial movements to support realistic actuation.

Development of Flexible Tactile Sensors Dec 2025 - Present

- Developing tactile sensors for robotic perception under Dr. Tushar Sakorikar.
- Project is currently in literature review and ideation phase with plans to start fabrication by February 2026.

Leadership & Volunteering

Research Head - Electronics and Robotics Club (ERC), BITS Goa Apr 2025 - Present

- Guiding **75+** students across multiple robotics projects (Vulcan, Snakebot, Quadruped)
- Managing project progress and integrating electronics, mechanical, and automation verticals.
- Organized and designed ROS2-based hackathon with **200+ participants nationwide** testing navigation and CV for warehouse robotics.

Committee Member - Sandbox Innovation Lab, BITS Goa Feb 2024 - Present

Maintaining lab equipment of the campus student-run (3D printers, laser cutters) and managing tech club purchases.

Volunteer - Abhigyaan, BITS Goa Oct 2023 - Present

Provided evening tuition classes for K-12 children of campus non-teaching staff.